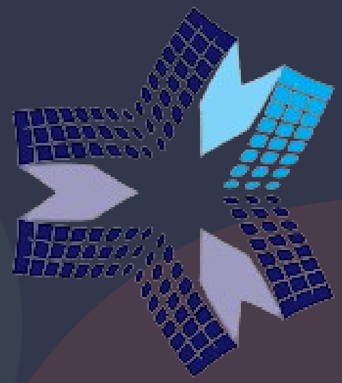
Tallinn, Estonia, 2026

Tsampodimos, Apostolos
FTMC



A Disformal Affair: How Dark Matter Mimics Lorentz Violation

G. Domènech, A. Ganz, F. Kirk, N. Sherrill, and A. Tsampodimos : Phys.Rev.D 113 (2026)
Apparent Lorentz violation from disformally coupled ultralight dark matter

G. Domènech, A. Ganz and A. Tsampodimos : JCAP 02 (2026)
On the consistent disformal couplings to fermions

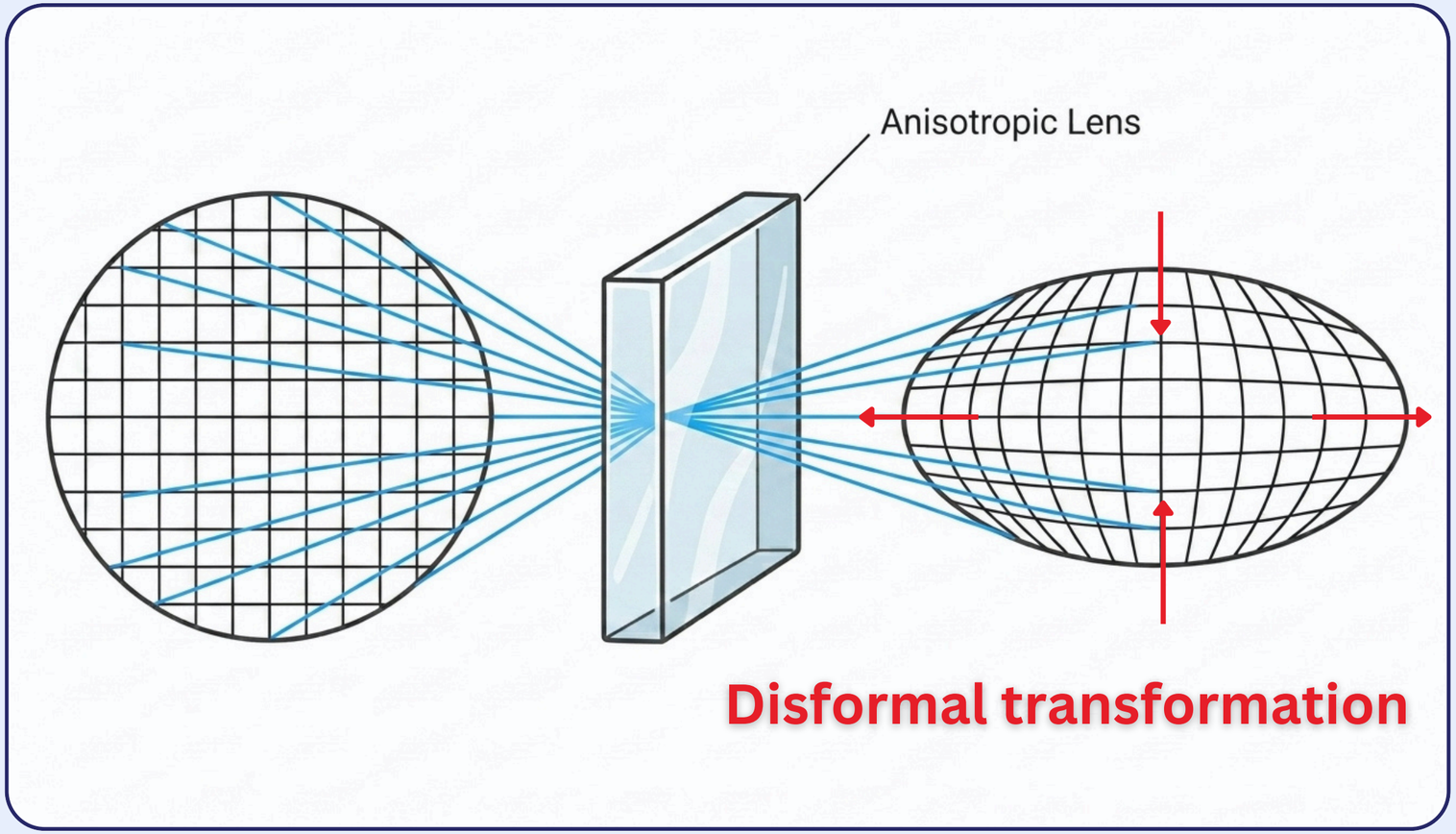
Disformal Transformations

*Bekenstein :

Gravitational metric g

Matter metric $\tilde{g}$

“Effective” metric



* J. D. Bekenstein : 10.1103/PhysRevD.48.364

Disformal Transformations

* **General disformal transformation**

$$\tilde{g}_{\mu\nu} = C^2(g_{\mu\nu} + 2L_{\mu\nu} + L_{\mu\sigma}L^{\sigma}{}_{\nu}) = C^2\mathcal{U}_{\mu}{}^{\sigma}\mathcal{U}_{\nu}{}^{\lambda}g_{\sigma\lambda}$$

Conformal **Disformal**

$$S = \int d^4x\sqrt{-\tilde{g}}\left[\frac{i}{2}(\overline{\tilde{\Psi}}\gamma^{\mu}\overset{\tilde{\leftrightarrow}}{\nabla}_{\mu}\tilde{\Psi}) - \overline{\tilde{\Psi}}m\tilde{\Psi}\right]$$

* A.Tsampodimos : Phys.Rev.D 113 (2026)

Acting on Spinors

$$S = \int d^4x \sqrt{-g}\left[\frac{i}{2}(\mathcal{U}^{-1})^{\mu}{}_{\nu}(\bar{\Psi}\gamma^{\nu}\overset{\leftrightarrow}{\nabla}_{\mu}\Psi) - \bar{\Psi}Cm\Psi - M_{\mu}J_5^{\mu}\right]$$

I. Super Simple

Simpler than bosonic:

- shifts kinetic term
- introduces axial term
- rescales the mass

Acting on Spinors

$$S = \int d^4x \sqrt{-g}\left[\frac{i}{2}(\mathcal{U}^{-1})^{\mu}{}_{\nu}(\bar{\Psi}\gamma^{\nu}\overleftrightarrow{\nabla}_{\mu}\Psi) - \bar{\Psi}Cm\Psi - M_{\mu}J_5^{\mu}\right]$$

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II. M-Axial term

- Spacetime vorticity, and potentially unstable

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I. Super Simple

Simpler than bosonic:

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II. M-Axial term

- Spacetime vorticity, and potentially unstable

III. SME-coefficients

Identify the correspondence

$$c_{\mu\nu} \leftrightarrow (\mathcal{U}^{-1})_{\mu\nu}$$
$$b_{\mu} \leftrightarrow -M_{\mu}$$

to the SME coefficients.

Apparent Lorentz Violation

Apparent Lorentz Violation

Example: $L^{\mu}{}_{\nu} = DB^{\mu}B_{\nu}$

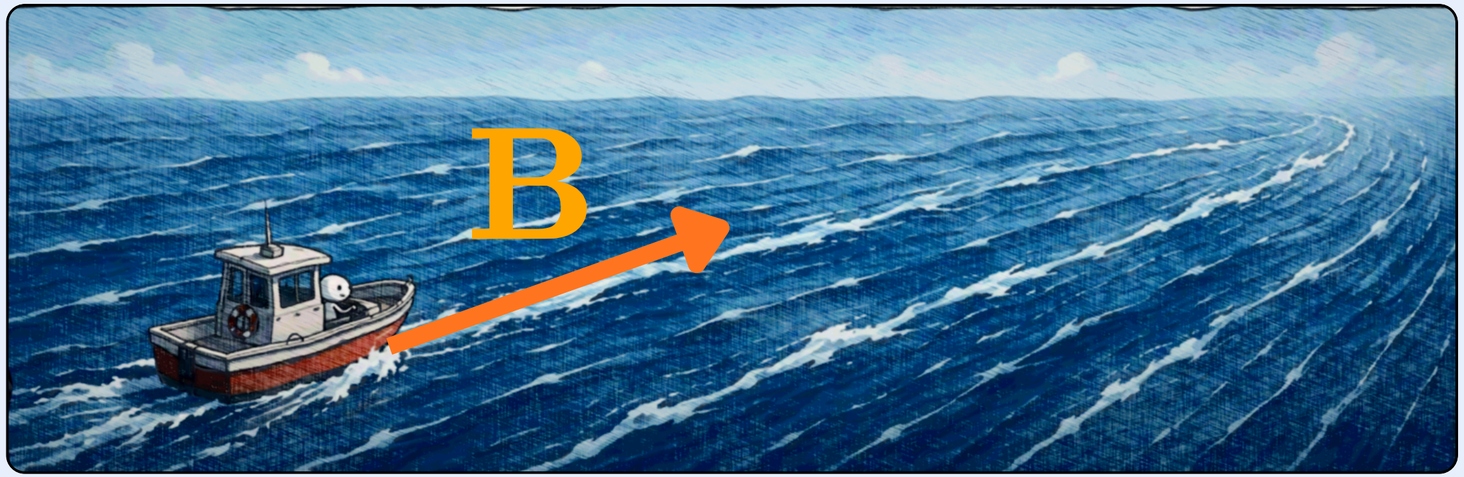
Could be vector dark matter

$$S = \int d^4x\sqrt{-g}\Big[\frac{i}{2}\bar{\Psi}\gamma^A\overset{\leftrightarrow}{\nabla}_A\Psi - \bar{\Psi}Cm\Psi - \frac{i}{2}\frac{D}{1+DX}B^AB_B\,\bar{\Psi}\gamma^B\overset{\leftrightarrow}{\nabla}_A\Psi + \frac{D^2X}{4(1+DX)}B^D\mathcal{F}^{\star}_{AD}J_5^A\Big]$$

Apparent Lorentz Violation

Example: $L^{\mu}{}_{\nu} = DB^{\mu}B_{\nu}$

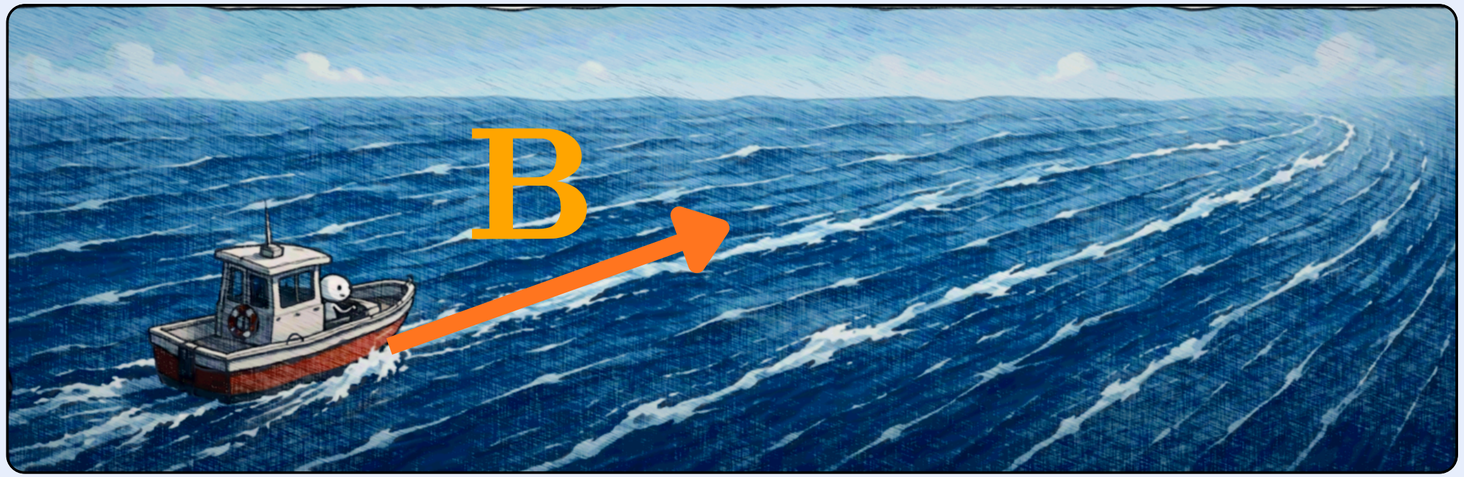
Could be vector dark matter

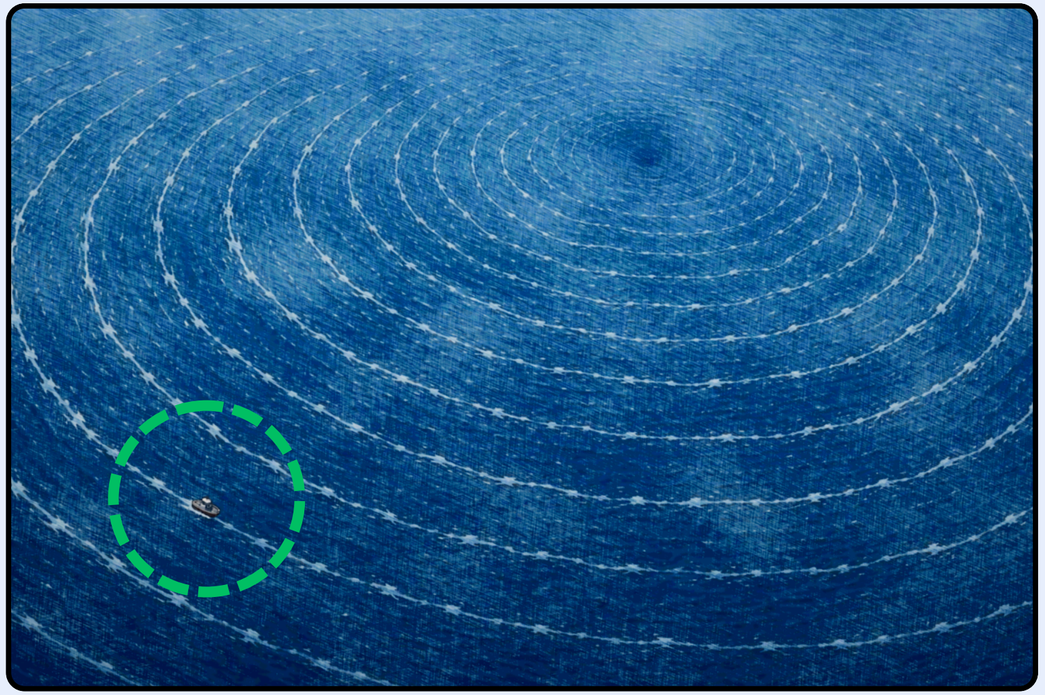


Apparent Lorentz Violation

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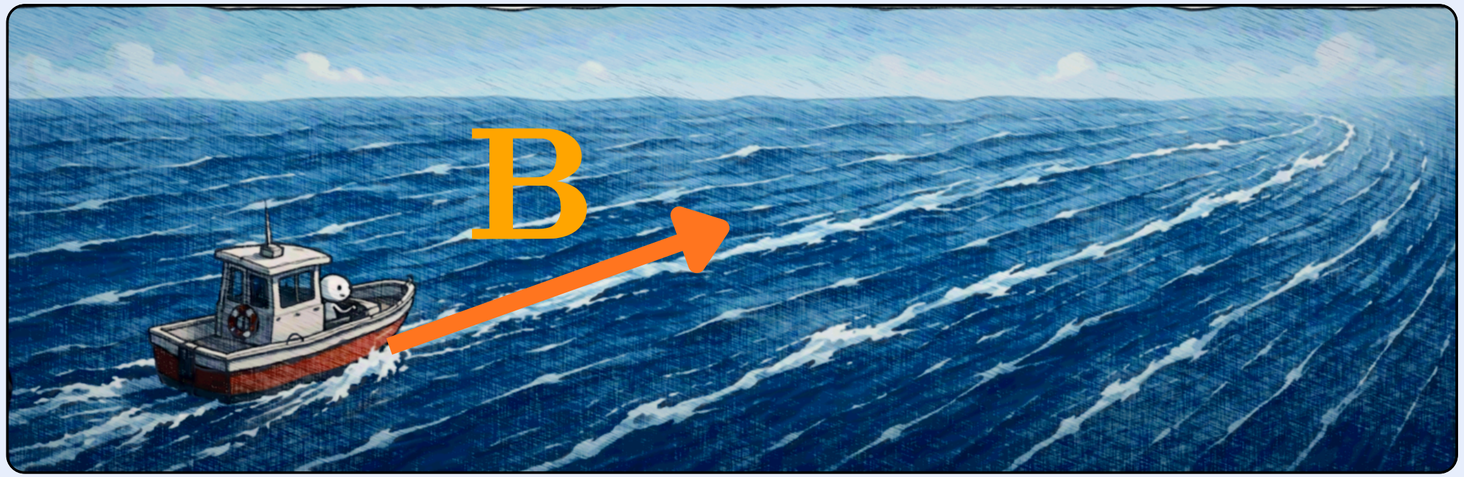


Apparent Lorentz Violation

Example: $L^{\mu}{}_{\nu} = DB^{\mu}B_{\nu}$

ULDM

Could be vector dark matter



Polarized Case

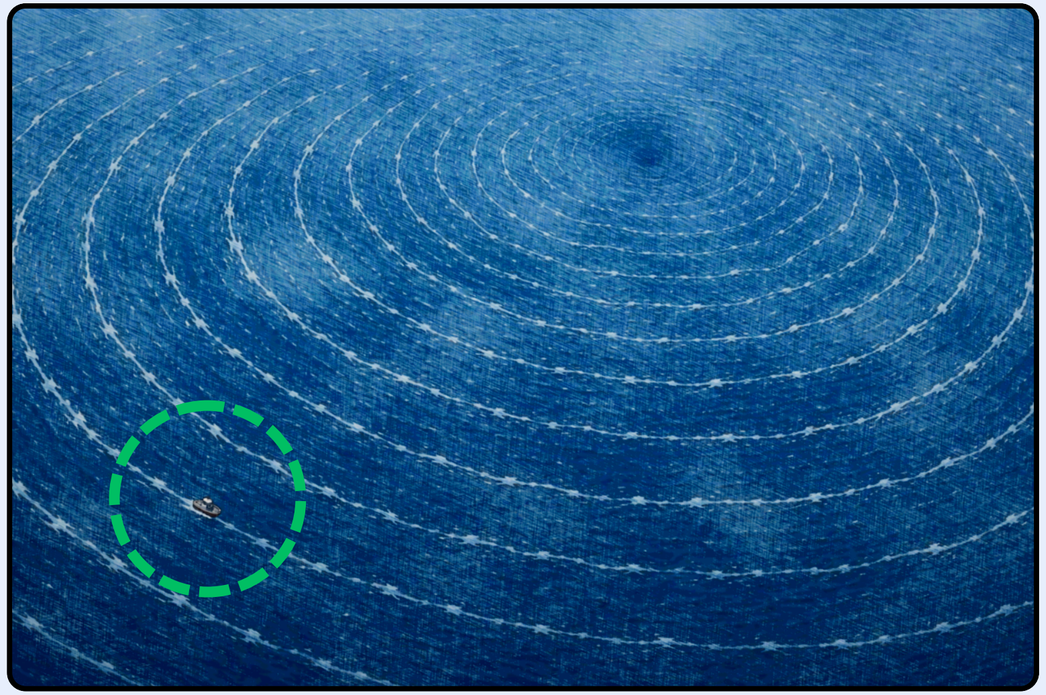
$$D \lesssim 6 \cdot 10^{-35}\, GeV^{-2} \left(\frac{m_B}{10^{-15} eV}\right)^2$$

$$\Lambda \gtrsim 10^{17}\, GeV \left(\frac{m_B}{10^{-15} eV}\right)^{-1}$$

Unpolarized Case

$$D \lesssim 6 \cdot 10^{-20}\, GeV^{-2} \left(\frac{m_B}{10^{-15} eV}\right)^2$$

$$\Lambda \gtrsim 10^{10}\, GeV \left(\frac{m_B}{10^{-15} eV}\right)^{-1}$$



Conclusions

1. General disformal transformation.
2. Bounds via particle physics data

Thank You

Bibliography



1. G. Domènech, A. Ganz, F. Kirk, N. Sherrill, and A. Tsabodimos : Phys.Rev.D 113 (2026)
2. *Apparent Lorentz violation from disformally coupled ultralight dark matter*
3. G. Domènech, A. Ganz and A. Tsabodimos : JCAP 02 (2026)
   On the consistent disformal couplings to fermions
4. J. D. Bekenstein : 10.1103/PhysRevD.48.364
   Relation between physical and gravitational geometry
5. M. Smiciklas, J. M. Brown, L.W. Cheuk and M. V. Romalis : PhysRevLett.107.171604
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